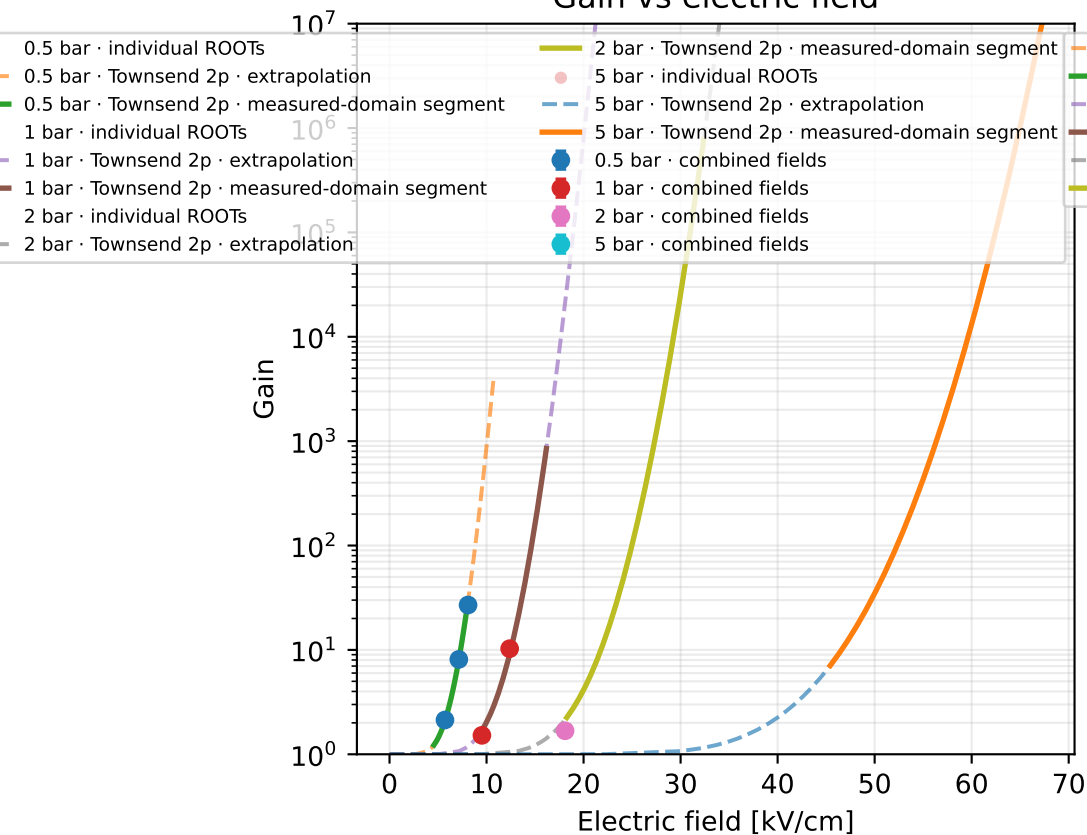
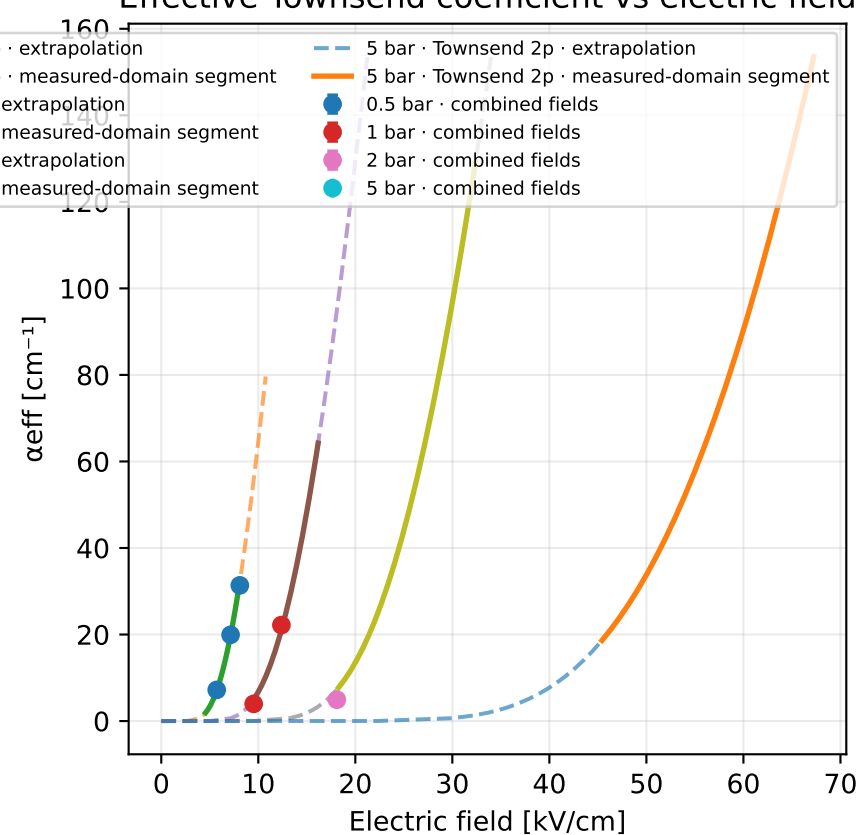


ArCF4 · Ar 99 % / CF4 1 % · gap 1.05 mm

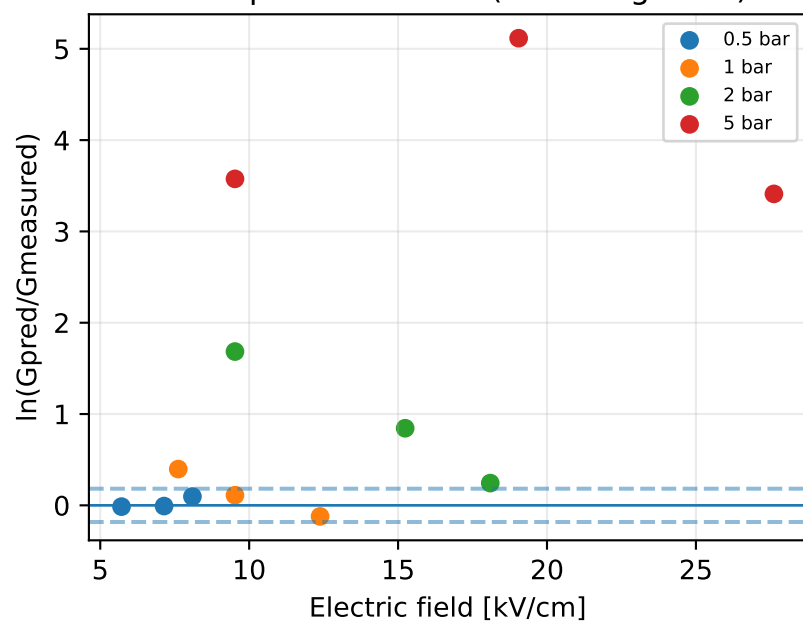
Gain vs electric field



Effective Townsend coefficient vs electric field



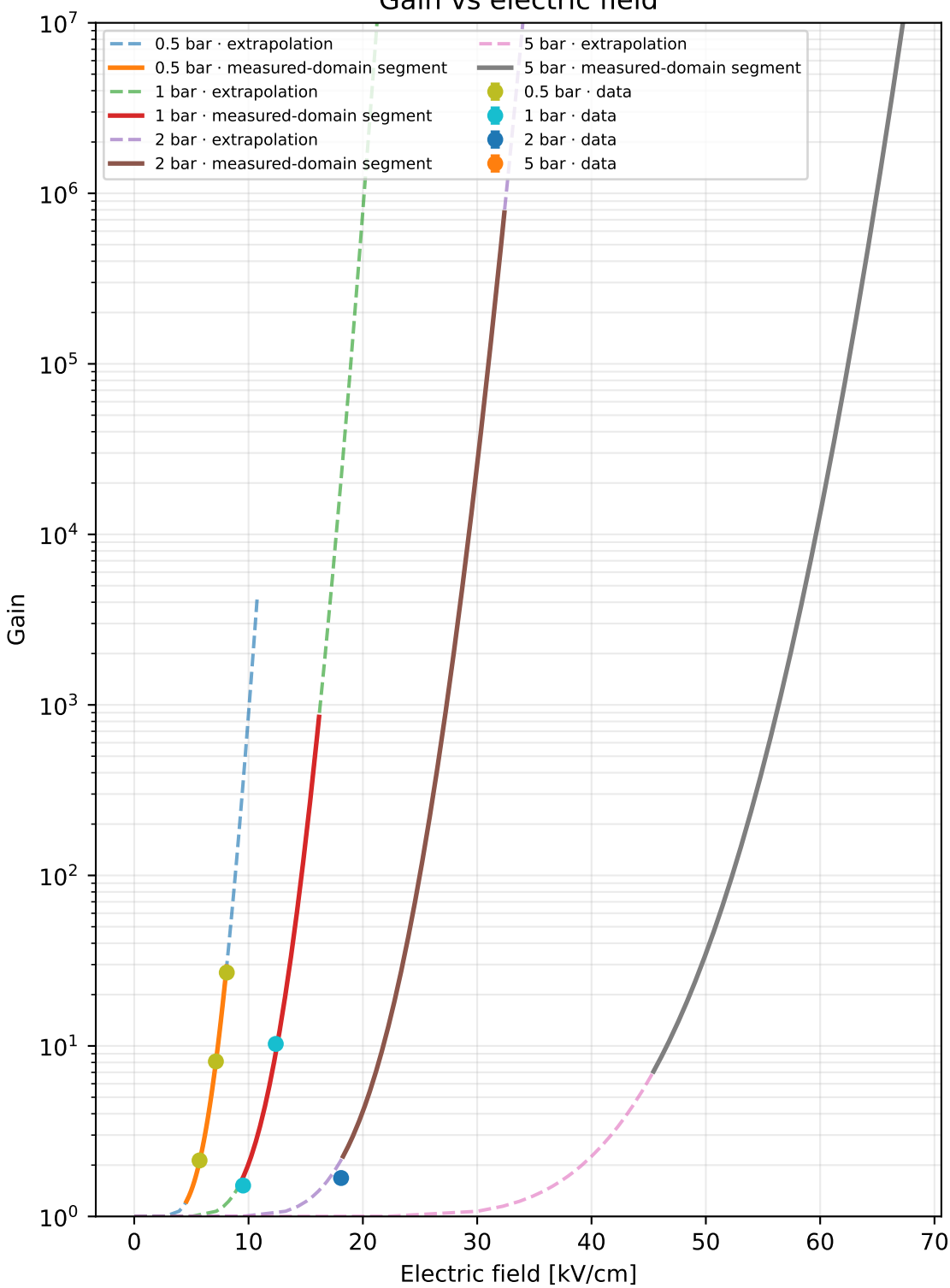
Gain-space residuals ($\pm \ln 1.2$ guides)



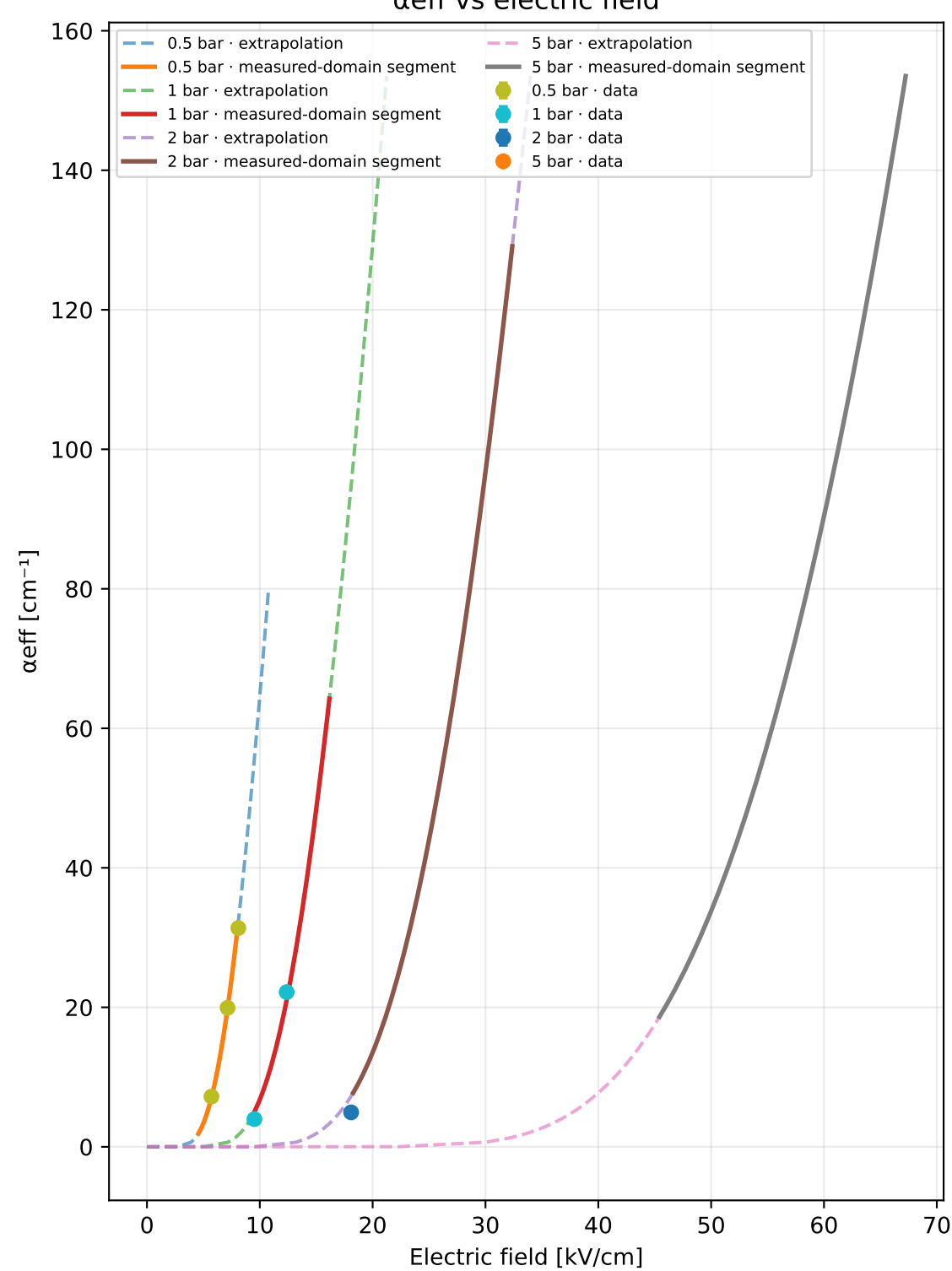
ROOT runs: 12
 Unique physical points: 12
 Pressures: 0.5, 1, 2, 5 bar
 Selected model: Townsend 2p
 Status: accepted
 Training RMSE: $\times 1.205$
 Cross-validation median: $\times 1.220$
 Cross-validation worst: $\times 1.503$
 Reduced-field range: 9.048–16.19 kV cm^{-1} bar $^{-1}$
 A=2474.4, B=59.03, m=0, n=1
 Covariance condition: 1.49e+03
 0.5 bar display limit: G=4.29e+03 (local inversion limit)
 1 bar display limit: G=1e+07 (requested cap)
 2 bar display limit: G=1e+07 (requested cap)
 5 bar display limit: G=1e+07 (requested cap)

Extrapolation capped at $G=1e+07$ or at the local model limit

Gain vs electric field



α_{eff} vs electric field



p [bar]	E [kV/cm]	E/p	Gain	σ Gain	α_{eff} [cm ⁻¹]	npe	runs
0.5	5.71429	11.4286	2.13	0.0876	7.20116	500	1
0.5	7.14286	14.2857	8.108	0.344	19.9319	500	1
0.5	8.09524	16.1905	26.93	1.07	31.3642	500	1
1	7.61905	7.61905	0.752	0.0325	—	500	1
1	9.52381	9.52381	1.518	0.0723	3.97518	500	1
1	12.381	12.381	10.272	0.481	22.185	500	1
2	9.52381	4.7619	0.186	0.0174	—	500	1
2	15.2381	7.61905	0.538	0.0378	—	500	1
2	18.0952	9.04762	1.68	0.102	4.94089	500	1
5	9.52381	1.90476	0.028	0.00739	—	500	1
5	19.0476	3.80952	0.006	0.00346	—	500	1
5	27.619	5.52381	0.034	0.00905	—	500	1

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-12.2	×1.22	×1.5	yes	—